

Emotion Recognition Using Speech Processing

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Abstract— Since many years ago, emotion identification from voice processing has been a significant factor in applications involving human-machine interfaces. In the psychological well-being of humans, emotions are pertinent. It serves as an agent for communicating one's viewpoint or mental condition to others. Speech Emotion Recognition (SER) is a manner of detecting the speaker's emotional state from the speech signal. Any computer system with limited processing resources may be programmed to sense or generate the few universal feelings, like Neutral, Anger, Happiness, and Sadness as needed. The following **characteristics** are retrieved in this work: Mel-frequency cepstral coefficients (MFCC), Chromogram, Mel scaled spectrogram, Spectral contrast, and Tonal Centroid. The emotion in this study is regarded using a deep neural network. The categorization of the speech in the output layer is done using Softmax. We used the combined 1440 audio recordings from 24 individuals in the RAVDASS speech dataset for training. Speech DNN has a 96% accuracy rate for emotion identification, which is greater than other algorithms like KNN, LDA, and SMO. Automatic emotion detection from human speech is becoming increasingly prevalent these days as it boosts interactions between computers and humans.

Keywords— Emotion recognition, Deep neural network, Audiofiles, Chromogram, spectrogram.

I. INTRODUCTION

Nowadays, many programmes make use of human-machine interaction. Speech is one of the interactional media. The primary challenge in human-machine tandem is a knack to understand feelings in speech. Furthermore, a variety of organic instructions can be used to assess emotion [1][2]. The main objective in this work is to identify emotions in speech. When two people engage with one another, they may quickly identify the underlying emotion in the other person's words. The intent of an emotion recognition system is to resemble how people see things [3]. Speech emotion recognition has several applications. Making decisions can be significantly influenced by emotions. A system can respond appropriately if emotion can be accurately detected in speech. Speech emotion recognition has several applications. Making decisions can be significantly influenced by feelings. A system can respond appropriately if emotion can be accurately detected in speech. Medical science, robotics engineering, contact center applications, and other fields may all gain from a robust emotion recognition system. [4][5]. Humans possess the ability of rapidly recognising speaker emotion. A great deal of practice and observation are required to accomplish this. Humans examine several aspects of an expression first, and then, via observations or prior experience, they identify the speaker's emotion.

Hence, it is necessary to develop an algorithm that is human-like and has the capability of accurately and quickly

detecting emotions. Multiple approaches are put forth in this discipline to identify a speaker's emotional state from the speech or voice signal. Anger, joy, sorrow, surprise, indifference, disgust, fear, tension, etc. are a few universal emotions. Numerous automated systems have been recommended by academicians during the past 20 years. The kind of characteristics that these various systems utilise to categorise speech signals vary. Mel-frequency cepstral coefficients (MFCC) are among the frequently utilised spectral characteristics. Our research revealed that the artificial neural network used to interpret speech emotion has a significant advantage throughout the recognition phase. Built-in emotion identification from human speech is becoming more prevalent nowadays as it improves links between both humans and machinery.

II. LITERATURE SURVEY

The authors[6] discuss using Deeply Sprint convolution neural networks (DSCNN) for the segmentation phase and the initial processing of speech signals to remove not beneficial noise and give clean voice. The goal is to create a voice emotion identification system employing an unaltered spectrogram and updated DSCNN structure.

Numerous analyses of various CNN and LSTM-RNN models were done in [15]. When contrasting with alternative LSTM and RNN designs, the CNN architectures performed superior. There are multiple hypotheses offered in [3] for the spectra and phoneme-based categorization of emotions.

[13] demonstrated an implementation of an end-to-end deep neural network for emotion identification. The paralinguistic information found in the words was used to create convolutional and recurrent networks by the researchers. A vector of attributes was created from the activations of the last fully-connected hidden layer in [14], where the authors studied the extensive spectrum properties created by passing the spectrum images over AlexNet for linguistic emotion identification.

A novel approach for categorising phrases according to mood. The algorithm used categorical HMM and short time log frequency power coefficients (LFPC) to characterise the algorithms and speech signals, accordingly. This technique divided the emotions into six groups, then trained and tested a novel system on an individual dataset. LFPC is contrasted with the mel-frequency Cepstral coefficients (MFCC) and the linear regression forecast Cepstral variables (LPCC) in order to assess the effectiveness of the suggested strategy. The results show that an average and highest identification rate of 78% was attained. Additionally, data show that LFPC is a superior feature than conventional characteristics for emotion categorization [12].

Despite adhering to any cultural or semantic data, Rong et al. [1] proposed an ensemble random forest to trees (ERFTrees) technique featuring an excessive range of characteristics for emotion identification. This technique is used with tiny data sets that have a lot of characteristics. An investigation using an analysis of Chinese speakers' passionate statement was conducted to assess the suggested technique, and the findings show that it improved the velocity of emotion identification. Additionally, ERFTrees outperforms emerging ISOMap and well-known scale reduction techniques like PCA and complex scaling (MDS). Whereas the weakest performance only achieves 16% on 84 parameters with the natural information set, the greatest accuracy with 16 features for female dataset reached an optimal accurate score of 82.54%.

For the sorting issue of speech emotion recognition, Ayadi et al. [4] suggested a Gaussian mixture vector autoregressive (GMVAR) technique, that uses a blend of GMM and linear average values. The main concept behind GMVAR is its capacity to distribute information across several media and construct a system that depends on voice set attributes. The Berlin emotional dataset was utilised to assess GMVAR. The research result indicates that the rate of classification may approach 76%, compared to 67% for feedback neural network networks, 67% for k-NN, and 71% for HMM. In comparison to HMM, this technique has the benefit of adequately separating high from unconscious states with neutral feelings [4].

III. METHODOLOGY

Deep neural networks are used in the suggested modified technique for voice emotion detection. To gain access to data obtained from the recordings file, this technique combines Mel-frequency cepstral coefficients (MFCC), Chromogram, Mel scaled spectrogram, Spectral contrast, and Tonal Centroid characteristics. The design divides the sound of the remarks into 8 categories, including impartial peaceful, joyful, depressed, furious, afraid, disgusted, and astonished. To derive voice intensity characteristics, a 5-layer a complex network is trained using DNN.

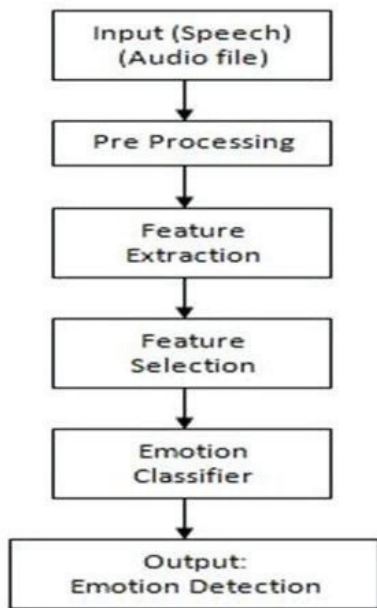


Fig. 1. Steps involved in speech emotion recognition

A. Data Collection:

The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) information set, being deposited in.wav file format, was used in this study. The librosa package in Python is used for accessing the files containing audio. The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) was utilised as the source data in this study. We picked the speech section, which has 24 performers (balanced by the concept of gender) and 1440 audio recordings.

B. Data Preprocessing:

Data preprocessing is a method for transforming unclear information into accurate data sets. The term "purifying the information." relates to the removal of voids, the replacement of empty strings with useful information, the removal of identical numbers, the removal of variations, and the removal of unneeded characteristics. Translate categorical variables to numerical values if the set of data comprises any information that fall into one of the categories. In sum, our model is 5 layers deep.

C. Feature Extraction:

Mel-frequency cepstral coefficients (MFCC), Chromogram, together with Spectral contrast and Tonal Centroid characteristics, are the features that were retrieved.

1) Mel-frequency cepstral coefficient

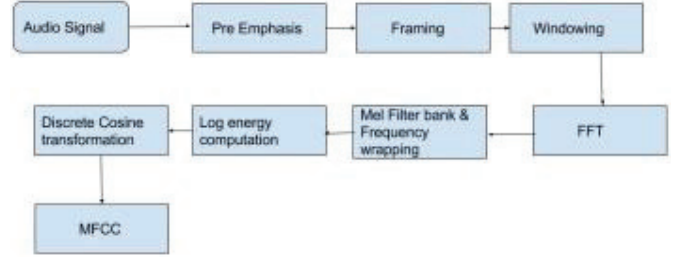


Fig. 2. BLOCK DIAGRAM OF MFCC

MFCC feature extraction process consists of a few steps as discussed below,

Pre-emphasis: For the indication to have more vitality, pre-emphasis is necessary. This procedure involves passing the spoken sample across an amplifier to boost its vibrancy. This increase in state of energy provides additional details.

$$K(z) = 1 - pz^{-1}$$

Framing: This step involves segmenting the voice sample into 20–40 ms frames. This procedure is required to set the size of speech because the voice of a person might fluctuate in duration. Although the voice message's essence is non-stationary—its amplitude might fluctuate across time—for a short amount of duration, it behaves as a fixed transmission.

Windowing: This step is carried out follows the frame procedure. The signal discontinuities at the beginning and conclusion of every picture are lessened by the windowing procedure. This method involves a 10ms phase transfer. This suggests half of the preceding pixel's content is repeated in every subsequent one.

Fast Fourier Transform (FFT): The frequency spectrum of each frame is produced using the rapid Fourier transform (FFT). The FFT converts every one of the samples of a frame between the duration phase to the frequency band. To

identify all frequencies present in an area, an FFT is utilised. Each image has 20–30 triangle filtration from the Mel scale filter bank implemented. The quantity of radiation is present in a particular picture is determined by the mel scale filter.

The following calculation can be used to translate the normal frequency f to the Mel scale m :

$$\text{Mel}(f) = 1125 \cdot \ln(1 + f/700)$$

Calculation of log energy: The log function is applied to the filtering branch energies of every frame once they have been obtained. Mammalian sense of hearing is another source of inspiration. Hearing in humans cannot function on a linear scale. The ears of humans is not capable of detecting significant electrical fluctuations when the audio level is great. Linear calculation of energy provides the qualities that a person can plainly hear.

Discrete Cosine Transformation (DCT): The logarithmic filtering bank values are used to compute the DCT in the final stage. We utilised 10ms of scrolling and 25ms of frames. Additionally, 26 band pass filters were utilised. 13 MFCC were calculated for every image. We calculated 13 MFCC characteristics from every frame. We additionally estimated the power contained in the frame itself. We estimated 13 velocity components and 13 acceleration components by calculating the time variations from energy and MFCC while obtaining 13 MFCC characteristics.

Generate the delta characteristics according to the previous and subsequent M structures, where $C_m(t)$ is the static coefficient of the panel.

$$DCT_{u,v} := G_{u,v} \cdot \sum_{n=0}^{N-1} \sum_{p=0}^{P-1} \left(M_{n,p} \cdot \cos\left((2n+1) \frac{u \cdot \pi}{2N}\right) \cdot \cos\left((2p+1) \frac{v \cdot \pi}{2P}\right) \right)$$

Where $C_m(t)$ denotes static coefficient of the frame, calculate delta features based on preceding and following M frames.

2) Spectral Contrast

The spectral contrast in a sound measures its **harmonic power at each epoch**. Because the majority of recordings of sound include a tone whose energy varies as time passes.

It seems challenging to gauge the vibration condition. This energy fluctuation may be measured using spectral contrast. **Simple, widen-band impulses often have significant contrast amounts, while broad-band signals typically have low contrast values.** Broad-band sound is typically associated with poor contrast values, although elevated contrast values typically correlate to distinct, shorten-band impulses. Here, the difference between the mean energy in the peak energy frame and the bottom or flat power frame is used to determine the power difference.

3) Tonal Centroid

The tonal centroids helps in Detecting Harmonic Change in Musical Audio or variances due to tones in audio.

$$\text{Centroid} = \frac{\sum_{n=0}^{N-1} f(n)x(n)}{\sum_{n=0}^{N-1} x(n)}$$

The following characteristics of the tonal centroid are included:

4) Frequency And Pitch

Pitch and amplitude Resonance and its equivalent subjective property pitch make up the most fundamental tonality-related signal component. Another of the primary fundamental qualities of vibrations, in addition to sound volume, length, the tone, and geographic position, is tone. A basic waveform is represented by a sinusoidal operate, $x(t)$, whereby a is the highest magnitude, t is the duration (in seconds), f is the frequency, and ϕ is the beginning phase. The amount of cycles that are replicated each second is known as the frequency f of this straightforward pattern.

5) Melody

A series of frequency sounds is typically thought of as a melody.

According to this explanation, a melody is "a collection of pitches of pitched sounds arranged in musical time in accordance to particular social norms and restrictions."

6) Harmony

The phrase "harmony" refers to the instantaneous pairing of sounds, known as chords, and over time, chord progressions. By addition to describing pitches and keys, the word primarily refers to a set of mathematical rules guiding how they are combined. Harmony, in the second meaning, has an array of concept books. We just took note of the components of the harmonic content which are to do with how pitches are put together into harmonies and how it relates to the item's tonality. As we've observed, rhythm and tune generally relate to a group of notes that are potentially synchronous or sequentially. Because they interact, it can be challenging to distinguish between harmony and melody in this manner.

7) Tonality

The ambiguity of the concept of tonality is one learning the causes of an absence of consensus across various methodologies for musicality induction. Castil-Blaze coined the phrase "tonality" for the first time in 1821. Today, it frequently refers to a set of interactions among a number of pitches that produce harmonies and melodies and include a tonic, or core pitch class, as its most significant (or permanent) component.

It describes the configurations of pitch phenomena in the broadest extent imaginable. The tonal centroids in order aid in harmonic change detection.

D. System Design:

1) Application for Emotion Recognition Model :

The user uploads an audio file that contains speech from a person, and the model is then utilised to forecast the speaker's emotional state.

The model is used in a web application built using the Flask architecture. The system and the user are its two components. The UI has a method for managing user login and authentication. Initially an unfamiliar user needs to enter their information, usually consists of name, email, and password. The database known as MySQL houses this individual's info. Someone who has already registered and whose information is recently in the system's MySQL database may access the internet-based application using authorised login details. They are only given permission to the programme to forecast feelings when they properly log in.

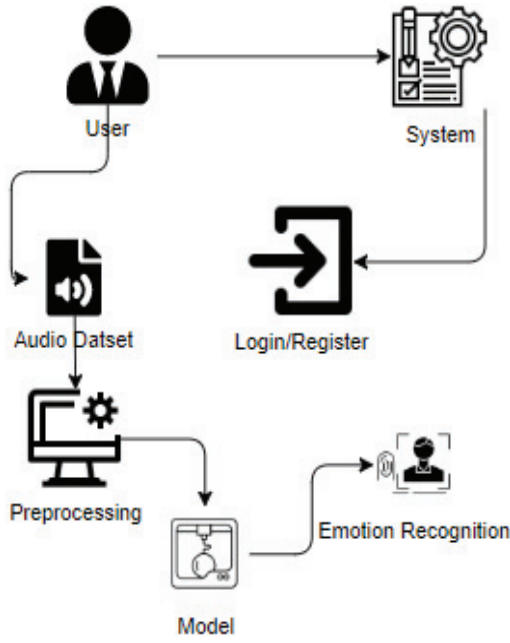


Fig. 3. Overall Workflow of the Emotion Recognition Application

E. Model for Deep Neural Network to Recognition Emotion:

Deep learning can boost efficiency while using fewer processors. In order to develop models for activities that typical machine learning algorithms cannot or are difficult to do, neural networks with deep connections (DNN) are frequently utilised in the field of deep learning.

1) Architecture Diagram of DNN

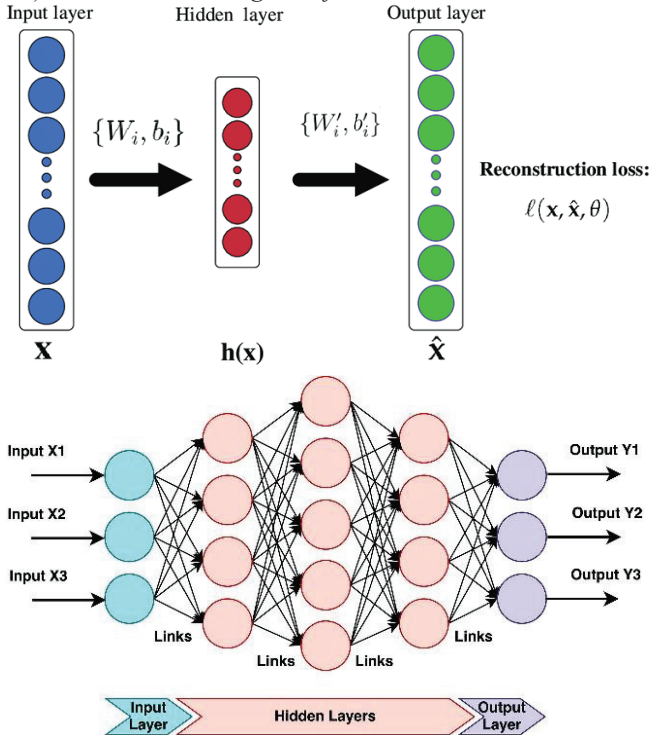


Fig. 4. Layers in Architecture diagram

Our 1440 audio recordings dataset was divided into training information (1008 audio files) and testing information (432 audio files). In this case, the initial dataset contains 80% of the data.

Five levels of a Sequence() model are produced. The set of data is trained employing 700 epochs using the train dataset. The model that we will use for prediction is the best in terms of accuracy when tested. We utilise softmax to classify the feelings in the output layer. Softmax is a accepts a vector of numbers and transforms them into chances, which are then applied to the output of picture generation.

F. Performance Metrics:

1) Hyper parameters

Classification rate

In the data from the test, the categorization ratio is calculated by summing the numbers on each diagonal and dividing the result by the sample size.

2) Recall

Recall is determined as the sum of true positives and false negatives weighted by the total amount of actual positives. Recall is defined as True Positives / (False Negatives + True Positives). The outcome is a number that ranges from 0.0 for no memory to 1.0 for complete or flawless recall.

3) Precision Value

Improving precision will reduce the amount of incorrect results, while improving recall will reduce the number of false negatives as well.

4) F-Measures

Recall and precision may be combined onto one metric using F-metric, which incorporates two attributes. F-Measure is equal to the product of the recall and accuracy factor.

5) ROC

Rate of Change (ROC) estimates the percentage change in price between the present price and the price after a certain amount of periods.

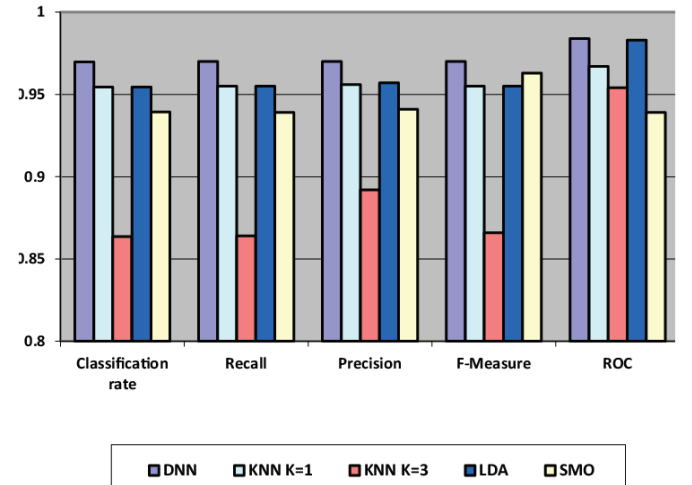


Fig. 5. Performance Metrics

G. Experiments And Result

We initially separated the entire set of information into two groups: 80% and 20%. Data were used for training purposes in the amount of 80% and for the purpose of validation in the amount of 20%. Then, for every file in the training dataset and test data set, we calculated the MFCC characteristics. We gave the deep neural network that processed the retrieved characteristics as its initial inputs. Using five convolutional layers, we utilise DNN. For the

network we operate, 700 epochs were established set. The final output nodes feature an activation function called "softmax" that we have employed. We also computed loss using classification linear sensitivity. Training performance peaked at 96% after 700 iterations, whereas test accuracy peaked at 80%. Confusion matrix for training data is represented graphically in Figures.

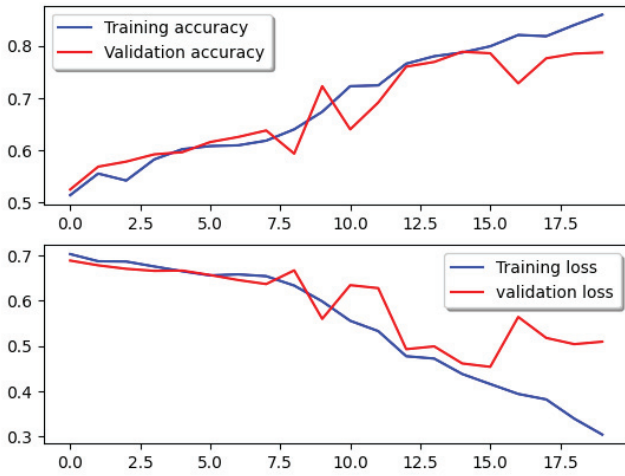


Fig. 6. Accuracy Graph and table

TABLE I. CONFUSION MATRIX FOR VALIDATION DATA

	Fear	Disgust	Happiness	Boredom	Neutral	Sadness	Anger
Fear	11	0	1	0	2	0	0
Disgust	3	7	0	0	0	0	0
Happiness	5	1	9	0	0	0	0
Boredom	1	0	0	14	2	0	0
Neutral	0	0	0	6	10	0	0
Sadness	0	0	0	1	0	12	0
Anger	0	0	1	0	0	0	25

The real recognition of emotion is represented by the confusion matrix's diagonal components. Our network can identify the correct emotion with a high degree of accuracy for the majority of emotions.

IV. CONCLUSIONS

Deep learning algorithms can produce effective outcomes. We have successfully created a deep learning model for emotion identification that scored 96% on tests. Please be aware that emotion prediction is arbitrary and that different people may grade the same music with different feelings. This is also the cause of the algorithm's

occasionally inconsistent output when trained on human-rated emotions. As the model was trained using the RAVDESS dataset, the speaker's accent might possibly provide unexpected results as the model is trained on North American accent database..

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